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Copper-Free One-Pot Sonogashira-Type Coupling for the Efficient Preparation of Symmetric Diarylalkyne Ligands for Metal-Organic Cages

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Abstract

Bipyridine- and benzimidazole-based ligands for the self-assembly of Co₄L₆ cages were synthesised in short reaction times and high isolated yields directly from aryl halide precursors using a copper-free one-pot Sonogashira-type coupling. This one-pot method circumvents the often time-consuming and challenging ligand synthesis for the preparation and application of cages.

Keywords

supramolecular chemistry, metal-organic cage, One-Pot, Sonogashira-type couplings, Ligand Synthesis

Copper-Free One-Pot Sonogashira-Type Coupling for the Efficient Preparation of Symmetric Diarylalkyne Ligands for Metal-Organic Cages

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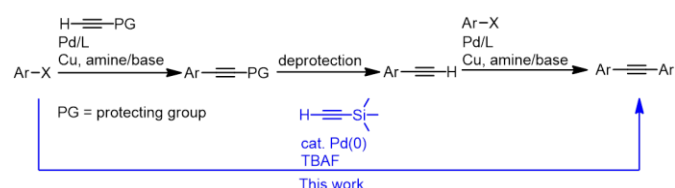
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Bipyridine- and benzimidazole-based ligands for the self-assembly of Co₄L₆ cages were synthesised in short reaction times and high isolated yields directly from aryl halide precursors using a copper-free one-pot Sonogashira-type coupling. This one-pot method circumvents the often time-consuming and challenging ligand synthesis for the preparation and application of cages.

Palladium catalysed cross-coupling reactions are powerful synthetic methods for carbon-carbon bond formation in modern organic chemistry.¹ In particular, Sonogashira cross-coupling reactions²⁻⁴ have been used extensively for the synthesis of diarylalkynes in natural products,⁵ conjugated oligomers/polymers in materials science^{6, 7} and ligands/building blocks⁸⁻²⁰ in coordination chemistry and supramolecular chemistry due to their typically high yields and tolerance of a wide range of functional groups.

Symmetric diarylalkynes find application in diverse fields since they are precursors in the synthesis of hexaarylbenzene derivatives,^{21, 22} building blocks for light-emitting materials²³ and inorganic heterocycles²⁴ as well as building blocks in supramolecular architectures.^{9, 10, 13} However, their preparation is typically a multi-step synthesis, often involving long reaction times and multiple purification steps, resulting in overall lower yields and making scale-up for applications difficult (black route, Scheme 1).²³ Glaser coupling²⁵⁻²⁷ can also take place in the presence of the copper co-catalyst with traces of oxygen, leading to side-product formation^{2, 28} and difficult purifications. As a result, one-pot cross-coupling methods using a variety of acetylene sources (e.g. gaseous acetylene,²⁹⁻³¹ calcium carbide,³² propiolic acid³³ and silyl-protected alkynes³⁴⁻³⁶) as well as copper-free Sonogashira-type couplings^{32, 33, 35, 37, 38} (e.g. replacing the copper co-catalyst and amine with tetrabutylammonium fluoride (TBAF)^{37, 38}) have been developed to overcome these problems, respectively.



Scheme 1. Multi-step synthesis of diarylacetylenes via sequential Sonogashira cross-coupling reactions (black) and the copper-free one-pot Sonogashira method (blue) in this work.

Ligand synthesis is a bottleneck for the self-assembly and application of metal-organic cages¹⁴⁻²⁰ due to multi-step syntheses and challenging purifications. Diarylalkyne-based ligands are appealing given their potential synthesis via a one-pot procedure and further functionalisation via the alkyne functionality, e.g. through post-assembly modification.¹⁰ However, most one-pot Sonogashira couplings have been reported for carbocyclic rather than heterocyclic diarylalkynes.³⁶

We report the efficient synthesis of symmetric diarylalkyne ligands **1a-c** and **2b-c** for metal-organic cages **3a-c** and **4a-b** (Figure 1) via a copper-free one-pot procedure using trimethylsilylacetylene as the acetylene source and TBAF functioning as a base, activator and deprotection reagent (blue route, Scheme 1). In addition to significantly reducing the synthetic burden from a 3 step synthesis with long overall reaction time to a single 3 hour step, the ligands were prepared in high isolated yields (32-92%) for a one-pot procedure. The proof-of-principle for large scale ligand synthesis was also demonstrated. Thus, this method enables rapid access to ligands for metal-organic cages from suitable aryl halide building blocks and this will facilitate the discovery of new cages as well as the translation of cages to applications.

We recently reported that the synthesis of ligand **1a**³⁹⁻⁴¹ could be reduced from three to two steps using TBAF for the *in situ* deprotection of the TMS protected alkyne (first step in black route, Scheme 2).¹³ However, the Glaser by-product was also obtained in the final Sonogashira coupling using copper(I) iodide in some instances. Mori³⁷ and Li³⁸ reported copper- and amine-free Sonogashira couplings between terminal alkynes and aryl halides, including aryl chlorides, with short reaction times and good to excellent yields using TBAF as an activator. It is proposed that the TBAF activates and stabilises the Pd(0) species, deprotonates the alkyne and acts as a phase-transfer catalyst.³⁸ Therefore, we envisaged TBAF could play the role of not only a deprotection reagent but also activator and base in a one-pot procedure while preventing the formation of Glaser by-products.

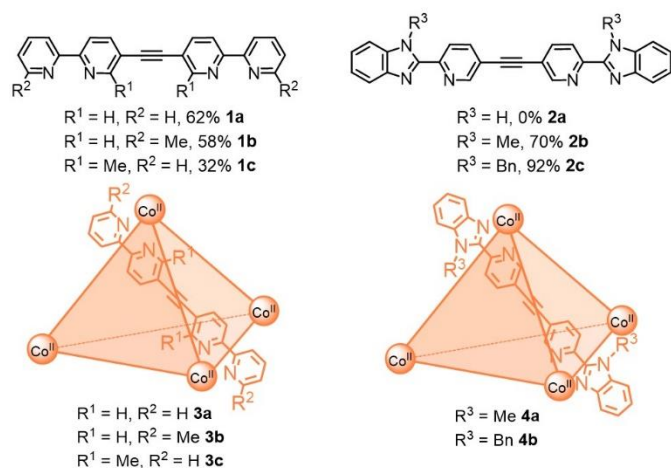
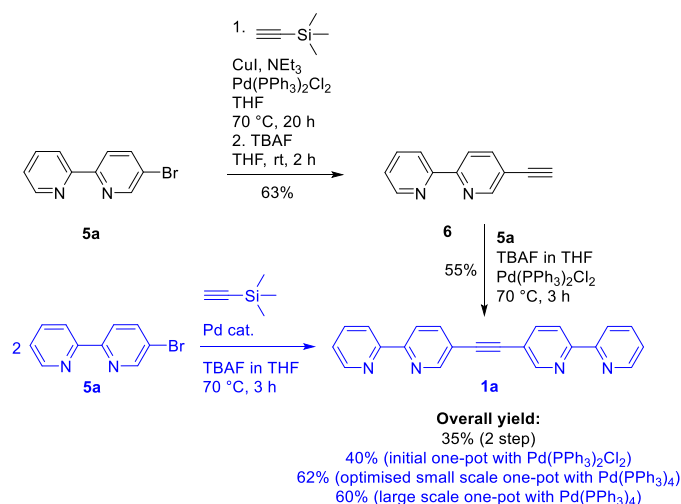


Figure 1. Symmetric heterocyclic diarylalkyne ligands synthesised using the copper-free one-pot Sonogashira-type coupling and their respective Co_4L_6 metal-organic cages.

Firstly, we investigated the applicability of the Sonogashira-type coupling using TBAF to heterocyclic diarylalkynes by reacting **5a** with **6**¹³ (second step in black route, Scheme 2). The previously reported Sonogashira coupling¹³ reaction conditions for **1a** were initially adapted replacing the copper(I) iodide and amine with a commercially available TBAF solution in THF. Since this would function as both a reagent and solvent, an excess of TBAF was used compared to the previously reported methods of Mori³⁷ and Li³⁸ to ensure both sufficient TBAF was present for its multiple roles in the reaction and a suitable reaction volume. Ligand **1a** was obtained with full conversion of **5a** within a shorter reaction time of 3 hours in 55% isolated yield following column chromatography (SI, Section 2.1.3.1). Taking into account the 63% yield for the synthesis of **6**,¹³ the overall yield of ligand **1a** from **5a** via the two sequential cross-couplings is 35% (Scheme 2).



Scheme 2. Development of the copper-free one-pot synthesis of ligand **1a** from **5a**: proof-of-principle for the use of TBAF as a deprotection reagent and activator in a 2 step synthesis (black route); yield and reaction time improvements through a one-pot approach and reaction optimisation (blue route).

Having demonstrated that TBAF functions in both the deprotection and Sonogashira-type steps (black route, Scheme 2), the one-pot synthesis of ligand **1a** was carried out under analogous conditions, using 1 eq. of **5a** and 0.5 eq. of TMSA (blue route, Scheme 2). Ligand **1a** was isolated in 40% yield and 12% of the starting material **5a** was also recovered (Table 1, entry 1). This with the observation of intense gas release upon addition of TBAF suggested immediate deprotection of TMSA's silyl protecting group, resulting in the release of gaseous acetylene. While no starting material was recovered upon increasing the TMSA to 1 eq., a similar yield was obtained (Table 1, entry 2).

Table 1. Initial one-pot Sonogashira-type experiments for the preparation of **1a** from **5a**.^a

entry	TMSA (eq.)	Reaction Vessel	Yield ^c (%)
1	0.5	RBF	40
2	1	RBF	35
3	0.5	PT	40
4	1	PT	53
5 ^b	1	PT	54

^a Reaction conditions: 1 eq. **5a** (723 μ mol), 5 mol% $Pd(PPh_3)_2Cl_2$, 6 eq. 1 M TBAF in THF (not degassed), N_2 atmosphere. RBF = round bottom flask, PT = pressure tube. ^b Degassed 1 M TBAF in THF. ^c Isolated yield after column chromatography.

The reaction vessel was then changed to a pressure tube to investigate if acetylene loss over the course of the reaction was significant. While the yield did not improve with a stoichiometric amount of TMSA (Table 1, entry 3), the yield could be improved to 53% by using an excess of TMSA (Table 1, entry 4). In entries 1-4 the reactions were carried out under a nitrogen atmosphere without degassing the TBAF solution in THF before addition to the reagents. Degassing of the TBAF solution by bubbling nitrogen was unnecessary as a similar yield was obtained when this additional step was performed (Table 1, entry 5).

In an effort to optimise the yield further, various palladium catalysts were screened (Table 2). Compared to $Pd(PPh_3)_2Cl_2$ (Table 2, entry 1), the catalysts $Pd(OAc)_2$ (Table 2, entry 2) and $Pd(dppf)Cl_2$ (Table 2, entry 3) gave lower yields. However, the reaction with $Pd(PPh_3)_4$ (Table 2, entry 4) gave the product in the improved yield of 62%. As a precaution, the TBAF solution for this reaction was degassed due to the oxygen sensitivity of the catalyst.⁴²

Table 2. Catalyst screening of one-pot Sonogashira-type experiments for the preparation of **1a** from **5a**.^a

entry	Pd catalyst	Temp. (°C)	Time (h)	Yield ^c (%)
1	$Pd(PPh_3)_2Cl_2$	70	3	53
2	$Pd(OAc)_2$	70	3	13
3	$Pd(dppf)Cl_2$	70	3	40
4 ^b	$Pd(PPh_3)_4$	70	3	62

^a Reaction conditions: 1 eq. **5a** (723 μ mol), 1 eq. TMSA, 5 mol% Pd catalyst, 6 eq. non-degassed 1 M TBAF in THF, pressure tube, N_2 atmosphere. ^b Degassed TBAF 1 M in THF. ^c Isolated yield after column chromatography.

Further experiments were carried out to investigate if the reaction parameters (time, temperature, catalytic loading, aryl halide reactivity) can be reduced without significantly impacting the yield (Table 3). With a 1 hour

instead of a 3 hour reaction time, ligand **1a** was still isolated in good yield and no starting material was recovered (Table 3, entry 2). However, reducing the reaction time further to 30 mins led to a significant decrease in the yield and 40% of the starting material was recovered (Table 3, entry 3). Although a shorter reaction time is possible, 3 hours was chosen for subsequent experiments to ensure full consumption of the starting material. Heating the reaction was necessary since the coupling did not take place at room temperature and 98% of the starting material was recovered, even with a longer reaction time of 20 hours (Table 3, entry 4). Although a ten-fold lower catalytic loading of 0.5 mol% still gave **1a** in 32% yield, the reaction was not complete within 3 hours since 41% of the starting material was recovered (Table 3, entry 5). Given the short reaction time with aryl bromide **1a**, the one-pot procedure was also extended to the less reactive aryl chloride **5d** (SI, Section S2.1.2) but no product was obtained (Scheme S2).

Table 3. Catalyst loading, temperature and reaction time screening of one-pot Sonogashira-type experiments for the preparation of **1a** from **5a**.^a

entry	Pd cat. (mol%)	Temp. (°C)	Time (h)	Yield ^b (%)
1	5	70	3	62
2	5	70	1	50
3	5	70	0.5	23 (40) ^c
4	5	rt	20	0 (98) ^c
5	0.5	70	3	32 (41) ^c

^a Reaction conditions: 1 eq. **5a** (723 μ mol), Pd(PPh₃)₄, 1 eq. TMSA, 6 eq. degassed 1 M TBAF in THF, pressure tube, N₂ atmosphere. ^b Isolated yield after column chromatography. ^c Recovered **5a**.

Through the development and optimisation of the copper-free one-pot procedure (SI, Section 2.1.3.2), ligand **1a** can now be prepared both more efficiently and without the formation of Glaser by-products. Li and co-authors proposed a mechanism for TBAF promoted cross-couplings³⁸ and we propose a similar mechanism for the one-pot procedure. Compared to the previous 3 step synthesis of ligand **1a** with a total reaction time over 40 hours,^{39, 40, 43, 44} this one-pot procedure gives the ligand in the significantly reduced overall reaction time of 3 hours. Furthermore, the overall isolated ligand yield has been significantly improved from 35% for the shortened 2 step synthesis (black route, Scheme 2) and 40% yield in initial one-pot experiments to 62% yield following optimisation (blue route, Scheme 2). This optimised yield corresponds to an average yield of 85% for each of the 3 steps (Scheme 1). Finally, the scalability of the procedure was investigated by increasing the reaction scale ten-fold to 1.7 g of **1a** (SI, Section 2.1.3.3). A similar isolated yield of 60% was obtained following a chromatography-free purification (Scheme 2).

Having optimised the one-pot conditions for the synthesis of **1a**, we extended the reaction scope to the related ligands **1b**, **1c** and **2a** based on 6-methyl-2,2'-bipyridine and benzimidazole coordination motifs, respectively (Figure 1). Ligand **1b** was synthesised in a comparable yield to **1a**, whereas **1c** was obtained in a lower yield, attributed to a steric clash of the methyl substituent in close proximity to the cross-coupling site

(Scheme S1). The synthesis of NH-benzimidazole **2a** was attempted using the same conditions (Scheme S5), however, only starting material was recovered. We proposed deprotonation of the imidazole under the basic conditions prevented the coupling taking place and therefore, the NH was substituted with a methyl (**6b**) or benzyl (**6c**) group (Scheme S5). Indeed, ligands **2b** and **2c** were isolated in good to excellent yields.[‡]

We previously reported the self-assembly of cage **3a**¹³ from **1a** and Co(NTf₂)₂ in CD₃CN. Self-assemblies with ligands **1b-c** and **2b-c** were investigated using analogous conditions; four equivalents of Co(NTf₂)₂ and six equivalents of the respective ligand were heated at 50 °C in CD₃CN for 20 hours. Cage **4b** was prepared *in situ* only, whereas cages **3b**, **3c** and **4a** were isolated by precipitation with diethyl ether. Characterisation using our recently reported paramagnetic NMR spectroscopy toolbox¹³ and mass spectrometry was consistent with tetrahedral Co₄L₆ cages (Figures S55-81). Further investigation of these cages and their properties will be the focus of future work.

In summary, we have reported a copper-free one-pot Sonogashira-type procedure for the preparation of symmetric heterocyclic diarylalkynes. We have applied this method to the synthesis of ligands **1a-c** and **2b-c**, which can be used to self-assemble a series of novel metal-organic cages **3b-c** and **4a-b**. This procedure has the following advantages: reduction of a 3 step synthesis to 1 step; good to excellent isolated yields (32-92%) for a one-pot synthesis; short reaction times of typically 3 hours; scalability; copper-free conditions preventing the formation of Glaser side-products; commonly used reagents that are commercially available and relatively inexpensive.

This one-pot procedure now enables rapid access to symmetric diarylalkynes ligands directly from aryl halide substrates both on a small and large scale. This reduces the synthetic burden for scale-up as well as incorporating additional functionality for applications. Libraries of ligands could also be generated for high-throughput screening of self-assembly conditions enabling the discovery of new cages. Given the application of symmetric diarylalkynes in numerous fields, this straightforward copper-free one-pot procedure could also be extended to other carbocyclic and heterocyclic aryl halide substrates and allow scale-up for applications.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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Notes and References

‡ Due to precipitation of the ligand from the reaction mixture, these ligands could be isolated without the need for purification by column chromatography.

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